

Take-home exam: December 18, 2014.

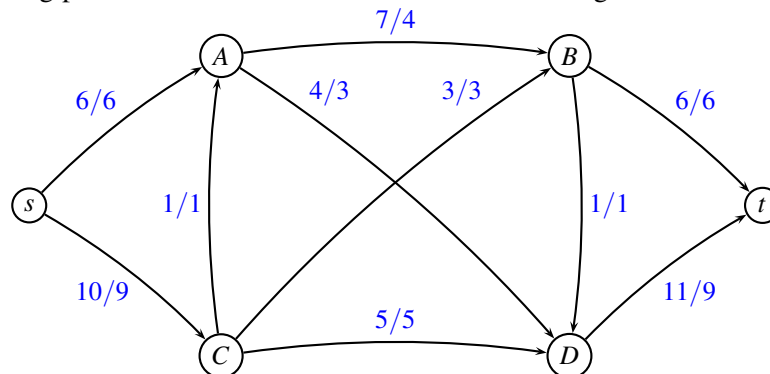
Course name: Network Optimization

Course no: 42115

Answer 1 Augmenting paths

iteration	augmenting path	flow
1	sABt	6
2	sCDt	5
3	sCBADt	3
4	sCABDt	1

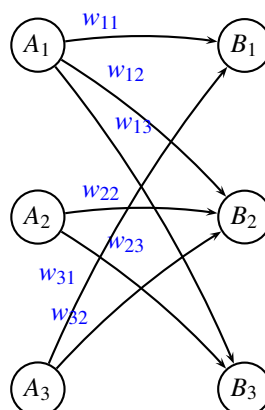
No more augmenting paths can be found. This leads to the following flow:



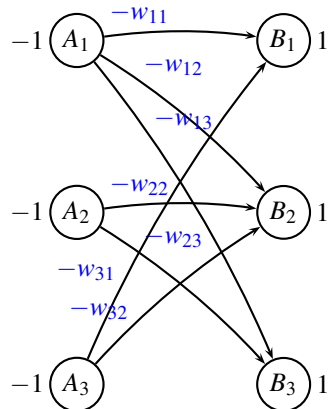
with objective value $z = 15$. ■

Answer 2 A minimum cut is found as $S = \{s, C\}$ and $T = \{A, B, D, t\}$. The capacity of the cut is $6 + 1 + 3 + 5 = 15$ which corresponds to the flow found above $z = 15$. ■

Answer 3 We have a bipartite graph, where the nodes can be split into two sets A and B , so that all edges (i, j) have $i \in A$ and $j \in B$. We may assume that $|A| = |B|$ since otherwise a perfect matching does not exist.



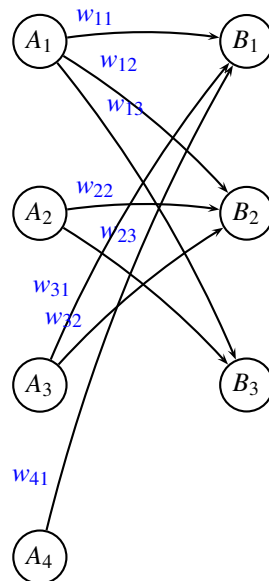
In order to transform the problem to a minimum cost flow problem, we set the demand to -1 for nodes in A , and to $+1$ for nodes in B . The weights are set to $-w_{ij}$ and the capacities are set to $u_{ij} = 1$.



A solution to the minimum cost flow problem will exactly find the perfect matching of largest weight.

■

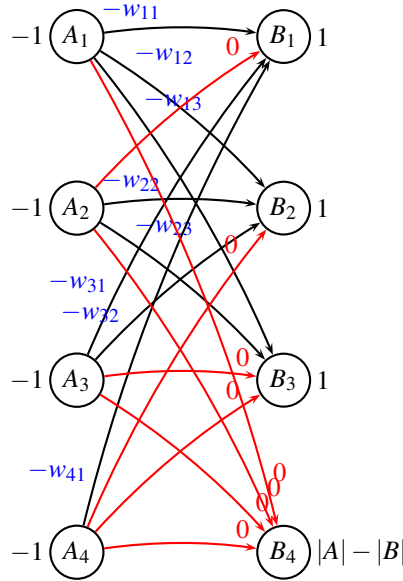
Answer 4 As before we have a bipartite graph, where the nodes can be split into two sets A and B , so that all edges (i, j) have $i \in A$ and $j \in B$. For symmetry reasons we may assume that $|A| \geq |B|$.



In order to transform the problem to a minimum cost flow problem, we set the demand to -1 for nodes in A , and to $+1$ for nodes in B . The weights are set to $-w_{ij}$. We introduce an extra node to the set B having demand $|A| - |B|$.

Finally, if two nodes i, j are *not* connected, we introduce a new edge (i, j) with weight $w_{ij} = 0$.

The capacities of all edges are set to $u_{ij} = 1$.



A solution to the minimum cost flow problem will exactly find a matching of largest weight. So, the problem can be solved in one run of a minimum cost flow algorithm. ■

Answer 5 Only model 5 can handle capacities on land. This is due to the fact that All transshipments between arrival of vessel H_i and vessel H_{i+1} share the same edge.

Model 3 and 4 have edges for land transportation, but we have direct edges between all pairs of vessels, so it is not possible to model a shared capacity.

Model 1 and 2 do not have any edges for land transportation. ■

Answer 6 In order to model tax costs, we need to have direct edges between all pairs of vessels.

Model 3 and 4 have direct edges between all pairs of vessels, so it is possible to model transshipment taxes.

Model 1, 2 and 5 do not have direct edges between vessels, so it is impossible to distinguish cargo from different vessels arriving at the port. ■

Answer 7 The reduced cost of a route r is calculated as

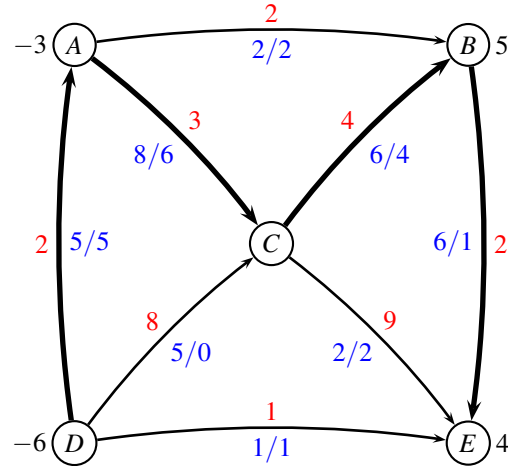
$$\bar{c}_r^k = d_k \sum_{(i,j) \in r} (c_{ij} - y_{ij}) - \bar{y}_k$$

Having the dual variables $y_{AB} = -2$, and $\bar{y}_1 = 14$, $\bar{y}_2 = 14$, $\bar{y}_3 = 6$, we get the following table:

number	commodity k	route r	d_k	$\sum c_{ij} - y_{ij}$	$\bar{y}_k$	reduced cost $\bar{c}_r^k$
1	1	E→A→B→D	2	7	14	0
2	1	E→D	2	7	14	0
3	2	D→C	2	8	14	2
4	2	D→E→A→B→C	2	7	14	0
5	3	E→A→B→C	1	6	6	0
6	3	E→A→B→D→C	1	15	6	9
7	3	E→A→D→B→C	1	5	6	-1
8	3	E→A→D→C	1	11	6	5
9	3	E→D→B→C	1	9	6	3
10	3	E→D→C	1	15	6	9

The most promising route to add to the model is route number 7 with reduced cost -1. ■

Answer 8 Every partitioning of edges in T, U, L defines a unique flow as shown in the notes. In the specific case we have



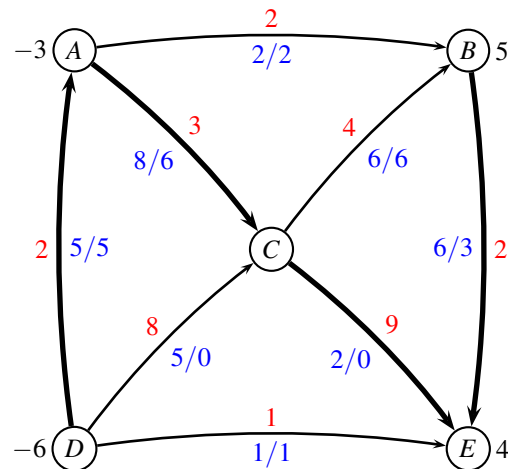
Notice that edges in T are allowed to be at the lower or upper bound as well as any value inbetween. The solution value is $z = 2 \cdot 2 + 6 \cdot 3 + 4 \cdot 4 + 5 \cdot 2 + 1 \cdot 2 + 2 \cdot 9 + 1 \cdot 1 = 69$ ■

Answer 9 Choosing D as root node, we find potentials (distances) $y_D = 0$, $y_A = 2$, $y_C = 5$, $y_B = 9$, $y_E = 11$. The reduced costs are

$$\begin{aligned}\bar{w}_{AB} &= 2 + 2 - 9 = -5 && U \text{ edge} \\ \bar{w}_{CE} &= 9 + 5 - 11 = 3 && U \text{ edge} \\ \bar{w}_{DC} &= 8 + 0 - 5 = 3 && L \text{ edge} \\ \bar{w}_{DE} &= 1 + 0 - 11 = -10 && U \text{ edge}\end{aligned}$$

Since edge (C, E) is a U edge and it has positive reduced cost, we can improve the solution. The edge (C, E) imposes the cycle E, C, B, E (we are going in the backward direction of edge (C, E) since it is a U edge).

The capacity we can send along the cycle is $\theta = \min\{2, 2, 5\} = 2$. Sending two items along cycle E, C, B means that edge (C, E) enters T while (C, B) enters U .

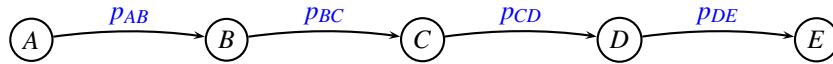


Choosing D as root node, we find potentials (distances) $y_D = 0$, $y_A = 2$, $y_C = 5$, $y_E = 14$, $y_B = 12$. The reduced costs are

$$\begin{aligned}
\bar{w}_{AB} &= 2 + 2 - 12 = -8 & U \text{ edge} \\
\bar{w}_{CB} &= 4 + 5 - 12 = -3 & U \text{ edge} \\
\bar{w}_{DC} &= 8 + 0 - 5 = 3 & L \text{ edge} \\
\bar{w}_{DE} &= 1 + 0 - 14 = -13 & U \text{ edge}
\end{aligned}$$

Since all U edges have negative reduced cost, and all L edges have positive reduced cost, we have found the optimal solution. The solution value is $z = 2 \cdot 2 + 6 \cdot 3 + 6 \cdot 4 + 5 \cdot 2 + 3 \cdot 2 + 1 \cdot 1 = 63$ ■

Answer 10 As an example consider a path $ABCDE$.



The probability for *not* being lost on the path is $(1 - p_{AB})(1 - p_{BC})(1 - p_{CD})(1 - p_{DE})$.

In general, for a path P going from s to t , we want to minimize the following expression

$$\max \prod_{(i,j) \in P} (1 - p_{ij})$$

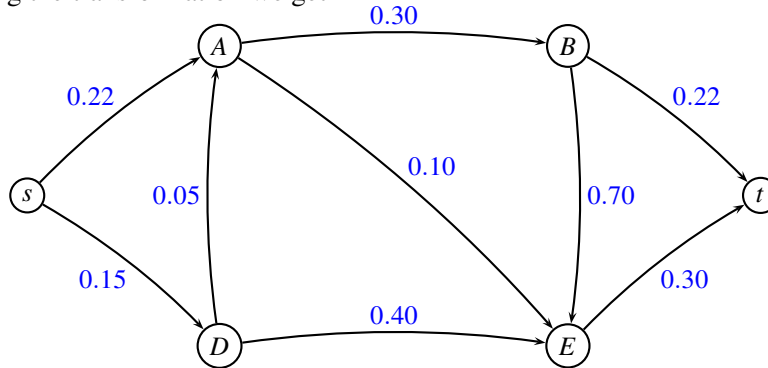
Taking the logarithm to this expression we can use the objective

$$\max \log \prod_{(i,j) \in P} (1 - p_{ij}) = \max \sum_{(i,j) \in P} \log(1 - p_{ij}) = \min \sum_{(i,j) \in P} -\log(1 - p_{ij})$$

So, using $w_{ij} = -\log(1 - p_{ij})$ as edge weights, the problem can be solved as a shortest path problem.

Notice that we can use the logarithm in the objective function $f(x)$, since the logarithm is a strictly growing function. This means that if $f(x_1) < f(x_2)$ then also $\log f(x_1) < \log f(x_2)$, so we will find the same minimum x , although the objective function $\log f(x)$ in the minimum will be different. ■

Answer 11 Using the transformation we get



■

Answer 12 Since $(1 - p_{ij}) \leq 1$ we have $\log(1 - p_{ij}) \leq 0$ and hence $w_{ij} = -\log(1 - p_{ij}) \geq 0$. We can therefore use Dijkstra's algorithm.

The following table shows the iterations:

considered node	potential					
	s	A	B	D	E	t
init	0	∞	∞	∞	∞	∞
s		0.22		0.15		
D		0.20			0.55	
A			0.50		0.30	
E						0.60
B						
t						

The shortest s - t path is therefore $sDAEt$ with cost 0.60.

To find the original probability of losing a message along the path $sDAEt$, we use the original probabilities. The probability for *not* losing a message is:

$$(1 - p_{sD})(1 - p_{DA})(1 - p_{AE})(1 - p_{Et}) = (1 - 0.3)(1 - 0.1)(1 - 0.2)(1 - 0.5) = 0.7 \cdot 0.9 \cdot 0.8 \cdot 0.5 = 0.252$$

So the probability for losing a message is around 75%. ■

CPLEX model for question 7:

```

minimize
+10x1+14x2+16x3+10x4+4x5+13x6+5x7+11x8+9x9+15x10
subject to
AB: +2x1          +2x4+1x5+1x6          <= 4
AD:              +1x7+1x8              <= 1
BC:              +2x4+1x5      +1x7      +1x9      <= 5
BD: +2x1          +1x6              <= 4
DB:              +1x7      +1x9      <= 1
DC:              +2x3          +1x6      +1x8      +1x10 <= 3
DE:              +2x4              <= 3
EA: +2x1          +2x4+1x5+1x6+1x7+1x8      <= 5
ED:      +2x2              +1x9+1x10 <= 3

K1: +x1+x2 >= 1
K2: +x3+x4 >= 1
K3: +x5+x6+x7+x8+x9+x10 >= 1

\\ R1: x1 <= 0
\\ R2: x2 <= 0
R3: x3 <= 0
\\ R4: x4 <= 0
\\ R5: x5 <= 0
R6: x6 <= 0
R7: x7 <= 0
R8: x8 <= 0
R9: x9 <= 0
R10: x10 <= 0
end

CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 18 rows and 10 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec.

Dual simplex - Optimal: Objective = 2.600000000000e+01
Solution time = 0.00 sec. Iterations = 0 (0)

CPLEX> dis sol var -
Variable Name      Solution Value
x1                  0.500000
x2                  0.500000
x4                  1.000000
x5                  1.000000
All other variables in the range 1-10 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
AB                  -2.000000
K1                  14.000000
K2                  14.000000
K3                   6.000000
R7                  -1.000000
All other dual prices in the range 1-18 are 0.

```